

Project Proposal & Plan

NutriVision: AI-Driven Dietary Health Assistant

ALX4_SWD8_S1

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1 Project Proposal

1.1 Overview

NutriVision is a health and wellness Flutter application designed to help users manage their diet seamlessly. It leverages Google's Gemini API for advanced AI integrations, allowing users to analyze meal descriptions and images, track daily calories, and receive customized nutritional advice based on their physical profiles.

1.2 Objectives

- Calculate personalized daily caloric needs based on user physical attributes (BMR & TDEE).
- Automate dietary tracking using the Gemini Vision API for food image recognition ("Snap Your Plate").
- Offer healthy recipe alternatives tailored to user restrictions using advanced prompt engineering.
- Use search and discovery prompts to suggest healthy snack alternatives tailored to user restrictions.
- Provide an intuitive, data-driven dashboard to visualize user progress.

1.3 MVP Scope (5 Weeks)

The MVP will be delivered within 5 weeks, focusing on a robust Flutter application utilizing `GridView` and `Stack` layouts. It includes Firebase Cloud Firestore for real-time data syncing, robust state management to separate UI from business logic, and Dio with interceptors for secure external API fetching. AI features will rely on direct Gemini API calls with strict system instructions (in-context learning).

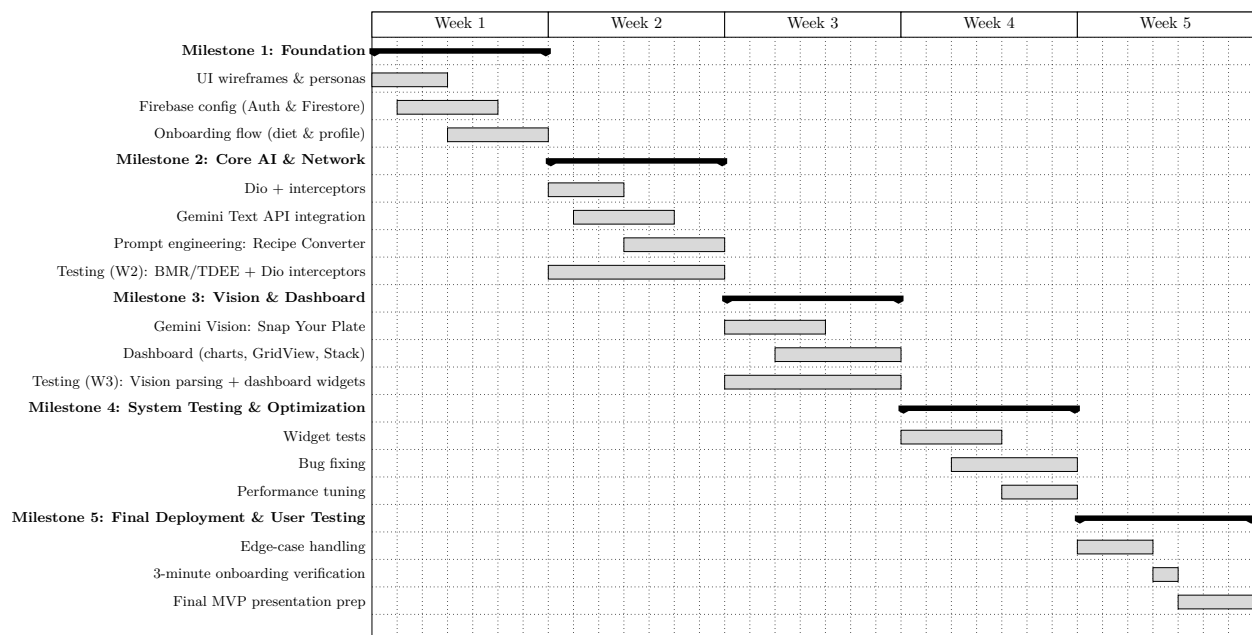
2 Future Scope & Scalability

While the current 5-week MVP relies on direct prompt engineering via the Gemini API, the long-term vision for NutriVision involves implementing a Retrieval-Augmented Generation (RAG) architecture using a custom Python backend (FastAPI) and a vector database (e.g., Pinecone). This future update will ensure 100% medical and nutritional accuracy by grounding AI responses in a verified local Egyptian and international food database, reducing the risk of AI hallucinations.

3 Project Plan

Week	Milestone	Deliverables
Week 1	Milestone 1	Foundation, UI/UX, & database setup: UI wireframes, user personas, Firebase configuration (Auth & Firestore base rules), and the user onboarding flow (diet & profile setup logic).
Week 2	Milestone 2	Core AI APIs & network integration: Dio setup with robust interceptors, Gemini Text API integration, prompt engineering for the “Recipe Converter” feature, plus snack discovery prompts; add unit tests for BMR/TDEE logic and Dio interceptor behavior.
Week 3	Milestone 3	Vision integration & data visualization: Gemini Vision API integration for “Snap Your Plate”, main interactive dashboard (using <code>fl_chart</code> , <code>GridView</code> , and <code>Stack</code>) to display calories and progress; add unit tests for Vision parsing and widget tests for dashboard components.
Week 4	Milestone 4	System Testing & Optimization: widget tests, bug fixing, and performance tuning.
Week 5	Milestone 5	Final Deployment & User Testing: edge-case handling, 3-minute onboarding verification (user testing), and final MVP presentation prep.

Gantt Chart (5 Weeks)



4 Task Assignment & Roles

To ensure a well-rounded experience, the 4 team members will divide work by features, while each member acts as a technical lead for a specific architecture layer:

1. **Marwan Mahmoud Abdelsalam (Auth & Profile Lead):** Setup Firebase Auth and Firestore; build the onboarding flow and diet calculation algorithms (BMR/TDEE); own and write unit tests for BMR/TDEE calculation logic and Firestore data persistence during the testing phase.
2. **Mario Nabil Samuel (AI & Prompt Engineering Lead):** Integrate Gemini Text API; design and optimize system prompts for the “Recipe Converter”; handle AI context logic; lead AI response validation testing, ensuring prompt outputs meet nutritional accuracy standards during the testing and refinement phase.
3. **Yusuf Mohamed Fawzy (Vision API & UI/UX Developer):** Lead the UI wireframes and user personas design in Week 1; develop “Snap Your Plate” using Gemini Vision API; ensure UI matches Figma wireframes and remains responsive across devices.
4. **Ahmed Mohamed Abdellatif (Network, Dashboard & Testing Lead):** Setup Dio with interceptors; build the analytical dashboard (charts, `GridView`); write unit/widget tests to ensure MVP stability.

5 Risk Assessment & Mitigation Plan

1. **AI API rate limits & network latency.**
Mitigation: Implement local state-management caching for frequent queries; use Dio interceptors for automatic retries, timeout handling, and user-friendly loading states.
2. **Inaccurate AI food recognition / hallucinations.**
Mitigation: Apply strict system prompting instructing the AI to rely only on general nutritional facts; provide a manual override so users can edit AI-generated calories before saving to Firestore.
3. **Team coordination and Git conflicts.**
Mitigation: Isolate features into specific Git branches (feature-branch workflow) and mandate code reviews before merging into the main branch.

6 KPIs

- **API response time:** AI responses and image processing resolve within 3–5 seconds.
- **User adoption rate:** Onboarding usability: the diet-profile onboarding flow is completed successfully in under 3 minutes during user testing sessions, with all profile and dietary preference data correctly persisted to Firestore.
- **System uptime & stability:** Zero critical crashes observed during the MVP demo and testing phase across all target devices (monitored via Firebase Crashlytics).

- **Testing coverage:** Minimum 50% unit-test coverage for core dietary logic (e.g., BMR/TDEE calculators) and critical data models, with all core calculation functions individually tested.